HOTEL MANAGEMENT SYSTEM

1 Introduction

The Hotel Management System is a tool for booking the rooms of Hotel through online process by the Customer. It provides the proper management tool and ease of access to the Customer information.

1.1 Purpose

The Hotel Management System's SRS' main objective is to provide a base for the foundation of the project. It gives us a view flow of "How the System Works?" and what the users' expectations are. The client's requirements are analyzed to build a desired system. This SRS for Hotel Management System can also be used on the basis for detail understanding on how the project was started.

1.2 Scope

The Hotel Management System project is inteded to make room reservation online. The system will have three end users:- Customer, Receptionist and Hotel Manager. It consists of the booking management system, database servers and report generators. Customers will be able to check room availability, select rooms and pay for them. Receptionist will have access to update or modify booking details. Manager can view the financial report and update the cost and category of rooms. The main goal is to simplify the everyday process of hotels. It will provide a web based interface accessible on desktops and mobile phones.

1.3 Overview

The document includes the general description of the system detailed functional requirements, design constraints, non-functional attributes and preliminary schedule and budget estimations.

2 General Description

2.1 Product Perspective

The Hotel Management System is a new self contained software product which will be produced by the project team in order to overcome the problems that have occured due to the manual system. The newly introduced system will provide an easy access to the system and it will contain user friendly functions with attractive interfaces. The system will give better options for the handling of large scale physical file system, errors occuring in calculations and all other required tasks specified by the client. The final outcome of this project will increase the efficiency of almost all the tasks done at the Hotel in a much convenient manner.

2.2 Product Features.

- Make Reservations
- Search Rooms
- Add Payment
- Issue Bills
- Manage Guest (Add, Update Guest)
- Manage Room Details (Add, Update, Delete)
- Manage Staff (Add, Update, Delete, View)
- Manage Inventory (Add, Edit, Delete)
- Set Rates

- Retrieve Reports (Staff, Payment, Income)
- Manage Users (Add, Update, Delete)
- Taking backups
- E-mail notifications.

2.3 User Classes and Characteristics

The user classes in Hotel Management System are:

- User
- Chef
- Room Attendant
- Financial Manager
- Owner
- Manager
- Receptionist

Characteristics of User Classes:

- User: The user can login into the system and book/cancel room reservations. He can also order food by looking up the menu in the system or ask for cleaning services. He can check in and check out virtually without the hassle of going to the front desk for everything. He can also make payment through various payment portals available in the system.
- Chef: The Chef gets notices whenever a User orders food and food preparations begin right away. After the food has been prepared, he notifies the room service for delivery of food so that the User's order reaches his room at the earliest.
- Room Attendant: Each Room Attendant is assigned multiple rooms and is notified whenever users post requests ranging from taking baggages from the reception to their room, cleaning the room, etc. They also deliver food from kitchen

to the User's room

- Financial Manger: The financial manager looks over the finances of the hotel right from booking the room to asking for food and User can either pay in advance or pay the lump sum at the time of Check-out.
- Owner: Hotel owner has the privilege of managing, monitoring and authorization of all tasks handled by the system. He can access every function performed by the system. Owner of the company as well as system can access the administration panel that is the core of the system. As the main authorized person of the company, the owner has the ability to mange the other users, their levels and privileges. System backups and restoration can be done by Owners. Meanwhile he will be able to take all the kinds of reports available in the system. As the owner of the system and the company, he has power to set room rates. He has the sole right of deleting a staff member from the database.
- Manager: Manager is responsible for managing resources available in hotel management system. Manager has most of the privileges mentioned above except payment handling. Using a manager reduces workload on the owner and assigns more responsible tasks to them that cannot be handled by the receptionist. At user level, manager has authority to take reports available in the system. Manager also has other abilities like adding staff to the system, modifying or removing them, adding new guests, adding new room types, etc...
- Receptionist: As the Receptionist, their role is to attain the goals of booking and ensure that all guests are treated with high standards of customer service. Hierarchically, this role has the least accessibility to system functions

after the Room Attendants. They play the boundary role in the system like registering new guests, making reservations, sending remainders to clients for booking confirmation, etc... Management will prefer to hire people who are efficient in communication and proficient in handling the electronic systems.

2.4 Design and Implementation Contraints.

Software development crew provides their best efforts in developing the system. In order to maintain the reliability and durability of system, some design and implementation constraints are applied. Some of them are:

- Memory: System will have minimum 10 GB of space data server.
- Language Requirement: Software must be only in English
- Budget Constraint: Due to limited budget, the system is intended to be very simple and just for basic functionalities.
- Reliability Requirements: System should sync frequently to backup server in order to avoid data loss during failure, so it can be recovered.

3 Requirements

3.1 Functional Requirements:

- Associate every online booking with an account.
- Limit every account to a single user.
- Enable users to search and find most relevant.
- Booking confirmation must be sent to specific contact details.
- Calculate and display accomodation charges.
- Cancel bookings
- Change rooms
- Order food
- Display and change record of guests.

3.2 Non Functional Requirements:

- Use encryption and captcha to avoid bots from booking.
- Search results should populate within acceptable time limits
- Users should be helped appropriately to fill in mandatory fields in case of invalid inputs.
- System should accept payment via various payment methods
- Easy to use, efficient and accessible.
- Keep track of documentation, activities and responses

3.3 Interface Requirements:

Hardware Requirements

- Operating System: Windows / Linux
- Computer: 4GB + RAM, monitor with minimum resolution of 1920 x 1080.
- Hard drive should be in NTFS file system formatted with minimum 10 GB of free space.
- Printer to print reports and notices.

Software Requirements

- Designed to run on any platform above Windows 7.
- Microsoft NET Frameworks 4.0 or above.
- Microsoft SQL Server Management Studio Express 2010.

3.4 Performance Requirements

- Data in database should be updated as soon as users register themselves
- Query of customer will be solved in time.
- Load time is less.
- Login process will be done quickly.

4 Preliminary Schedule and Budget.

4.1 Schedule

Week	Key Goals	Tasks	Milestone Achieved
Week 1	Requirement Elicitation	Conduct stakeholder interviews, define system scope, start SRS draft.	Requirement phase initiated.
Week 2	Requirement Finalization	Complete SRS, validate requirements with stakeholders.	Requirements finalized.
Week 3	System Design	Create architecture diagrams, database schema, select tech stack.	Design documentation ready.
Week-4	Start Development	Implement booking UI, initialize backend modules.	Frontend and backend scaffolding completed
Week-5	Continue	Develop API integrations, handle user auth, basic room logic	Basic system functions coded.
Week 6	Complete Development	Finalize business logic, integrate payment and notifications	Feature-complete version
Week 7	Integration	Link frontend with backend sync room and reservation modules	System integration successful.
Week 8	Testing Begins	Perform unit, functional and negative tests.	Initial testing completed.
Week 9	QA and bug fixing	Fix bugs, validate edge cases retest modules.	Initial testing completed.

Week	Key Goals	Tasks	Milestone Achieved
Week 10	Deployment and Docs	Deploy to staging/production write admin/user manuals.	System deployed.
Week 11	Buffer / Delays	Address any late bugs, test rollback scenarios, finalize	Contingency covered.

Detailed Explanation of Each Week's Tasks

- Week 1-2 Focused on requirement collection and documentatio This involved direct interaction with hotel stakeholders to understand real-world needs
- Week 3: System architecture was planned, including layered design, database schema and module responsibilities
- Week 4-6: Frontend UI components and backend logic for reservation, room management and payments were developed in parallel.
- Week 7: Frontend and backend were integrated. Room availability, booking flows and real-time updates were validated.
- Week 8-9: Thorough testing was conducted across modules. Edge cases and incorrect user actions were validated.
- Week 10: System was deployed to a working environment. Documentation was prepared for users and administration.
- Week 11: Reserved as buffer in case of unexpected delays or bugs.

Budget

Week	Key Goal / Phase	Estimated Cost
Week 1-2	Requirement Collection and Documentation	₹ 50,000
Week 3	System Architecture and Design	₹ 1,00,000
Week 4-6	Development Phase	₹ 3,20,000
Week 7	Integration	₹ 75,000
Week 8-9	Testing and QA	₹1,25,000
Week 10	Deployment and Documentation	₹62,000
Week 11	Buffer / Contingency	₹60,000

Grand Total: ₹7,17,000 (Estimation)

Notes:

- Costs include manpower only; licenses / hosting can be listed separately if needed.
- Week 11 serves as contingency buffer for risks/ overruns
- Adjust hours / rates per team reality the structure stays IEEE compliant (tracable to schedule phases and deliverables)

This SRS defines and provides a complete description of the Hotel Management System, covering objectives, functional, non functional, other requirements, schedule and budget. It serves as single reference document for developers, testers and stakeholders of project.

CREDIT CARD PROCESSING

Abstract

The Software Requirement Specification is designed to document and describe the agreement between the customer and developer regarding specification of software product requested. Credit card processing is interaction of customer with the credit card processor to buy their goods and pay bills. The processor handles card purchases by securely fetching card details and logging the transaction for processing.

Problem Statement

Credit card processing through offline involves the merchant collecting order information (including credit card numbers), storing this in their database and entering it using their on site credit card processing system. It takes time to manually enter credit card information for each order. It also creates other frauds like:

- Insecurity: There is a possibility that a skilled hacker could break into the database and steal an entire list of credit card numbers, thereby damaging the merchant's reputation with current clients.
- Higher risk of customer chargebacks: With no signature, there is higher risk of customer chargebacks.
- Higher risk of fraud: This comes into picture when people use stolen credit cards.
- Lack of trust: Many discerning online shoppers will not give their credit card to an "untrusted" online merchant.

Therefore, there is need for trusted and online credit card processing system.

1 Introduction

A credit card is a small plastic card issued to users as a system of payment. It allows its holder to buy goods and services based on the holder's promise to pay for these goods and services. The issuer of the card creates a revolving account and grants a line of credit to the consumer (or the user) from which the user can borrow money for payment to a merchant or as cash advance to the user.

When a purchase is made, the merchant swipes the card. The information goes to a gateway processor, which either accepts or rejects the transaction. If it is accepted, the transaction is held until the end of the business day. The merchant then re-enters the transaction via the gateway processor, the data is logged, and the debt is transferred to the account. The use of ATM for cash advance is a similar process.

If a business is selling to consumers, merchant services allow it to expand its customer base and provide a more convenient method of payment than cash or cheques. For online businesses accepting credit card processing becomes essential. Accepting credit cards allows funds to be transferred to the business' bank account in less than a week, which can be a welcoming relief for businesses experiencing tight cash flow.

The two purchase options for credit card processing facility are:

- Validation only: Credit card processing (which secures deposits at the time of booking)
- Credit Card Processing: With either options, credit card accounts entered during booking are validated to assure that the account is active and in good standing. The credit card processing option also allows properties to process credit card deposits.

1.1 Purpose

When customers complete their shopping cart, their credit card is preauthorized, and the order is entered into sales order. Credit card processor dials out and obtains a credit card payment. Within minutes, the customer receives receit in e-mail.

1.2 Scope

The credit card processing is usually done with the help of a swiper which scans all the details of card. After each purchase, the details are recorded and document is made by the credit card issuer.

Accepting credit cards is an integral part of businesses today for merchants who want to be competitive in their market and grow their businesses to its greatest potential.

Software and gateway processing, helps reduce fraud losses, saves you time and money, and provides powerful features and performance including detailed transactions' records and reports.

Swiping credit card ensures lower rates, resulting in potential saving of hundreds of dollars each month for businesses.

1.3 Overview

The SRS includes majorly four sections: overall description, specific requirements, schedule and budget.

Overall description will describe the major role of the system components and their interconnections.

Specific requirements will describe the roles and functions of the actors

The schedule will outline the planned timeline, key milestones and sequential tasks required to complete the system development

Budget will represent the estimated cost of development.

2 General Description (Overall Description)

2.1 Product Perspective

The Credit Card Processing System is a standalone software solution designed to handle card purchases, payments and transaction records efficiently. This system is developed to overcome the limitations of manual card processing, and minimize errors caused by human intervention.

The proposed system will allow users to make secure purchases, process payments and retrieve transaction details in real time. It will be designed to be user-friendly, with flexible options to handle different card types, and payment modes. The system will significantly reduce delays in payment processing, eliminate calculation errors and store a reliable copy of each transaction for future reference. The final outcome of this project will increase efficiency, accuracy, and convenience in card based transactions, by automating almost all tasks in a secured and structured manner.

2.2 Product Features / Functions

The system provides secure purchase processing and card management functionality.

- Front end: Enables users to initiate purchases, choose from multiple card types and select payment mode (one-time, EMI, split pay)
- Back end: Validates card details, preserves a detailed record of every purchase, processes settlements and generate transaction reports automatically.
- Security: Performs fraud detection and authorization checks to ensure only legitimate transactions are processed.
- Transaction Tracking: Maintains real time status of each purchase, including partial captures and refunds.

- Users Notifications: Sends instant alerts for approval, declines refunds, and payment confirmations.
- Audit and History: Provides complete transaction logs for compliance and reconcilation.

Major Use Cases:

- Authorization: Validates card type, payment mode and availability of funds before processing a purchase.
- Capture: Finalizes authorized transactions and sends funds securely to merchant.
- Refund (Credit): Allows full or partial reversal of a purchase if needed.
- Void: Cancels unsettled purchases before funds are transferred avoiding duplicate charges.
- Verification: Performs small-amount or zero amount checks to confirm customer identity and prevent fraudulent payments.
- Transaction Record Preservation: Stores detailed transaction history, enabling audits and customer support queries.

2.3 Users Classes and Characteristics

- Customer (User)
- Merchant
- Credit Card Bank (Issuer)
- Merchant's Bank (Acquirer)
- Authorization Service
- Deposit Processing System
- Reporting and Audit System

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Characteristics of User Classes

- Customer: They initiate purchase transactions or bill payments using a credit card. They provide card details, choose preferre-d payment modes, and receive instant notifications regarding approval, decline or refund.
- Merchant: Merchants submit transaction requests through payment gateway on behalf of customer. They rely on real time conformation of approval or rejection to finalize sales.
- Credit Card Bank (Issuer): The issued bank validates card details, available credit limits and performs fraud checks. It then approves or rejects the purchase request and generate-s an authorization code when successful.
- Merchant's Bank (Acquirer): The acquiring bank receives approved transaction data and ensures funds are deposited into merchant's account during settlement.
- Authorization Service: The authorization service validates the card number and expiry date while also authenticating the cardholder to prevent fraudulent transactions.
- Deposit Processing System: This component applies approved transaction amount to the customer's account and updates account balances accordingly.
- Reporting and Audit system: The report and audit system generates credit card transaction reports containing authoriza-tion codes, transaction amounts and success or errors messages. It maintains a record of all transactions for dispute resolution and compliance purposes.

2.4 Design and Implementation Constraints

The design and implementation constraints are:

- The system shall be accessible through any device with a stable internet connection and browsing capability.
- All user, product and indexing information shall be stored securely in a database ensuring 24 X 7 availability.
- The software shall be platform independent and run seamlessly on desktops and mobile phones.
- Users must login with valid credentials to access their accounts and perform actions.
- Payment processing shall be handled by a trusted third party processor offering fast fund transfers, low transaction fees, and fraud prevention measures suitable for high volume transactions.

3. Requirements

3.1 Functional Requirements

Administrators:

- Database Management: This should control the entire database and keep track of all records of customers and employee details. If there is any problem in the database (whether it is for accessing or updating) will be solved as possible without delay.
- Contact and Giving Permission to Vendors: Contact with vendors and give permission to sell their product under the site. Quality should be checked for first time before selling, after that the quality can be assessed by ratings of customers so that further processes can happen
- Details: Able to view details of all employees and control their details.

Customers:

- Registration: New users should sign up by creating new ID with either mobile number or mail account which is mandatory for placing order.
- Login: Customers must have valid credentials to enter into site.
- Details: They can view/edit personal, payment details, etc...
- Choosing and Comparing Products: Can view all available and similar products and compare them and make choices for purchasing
- Purchasing: Can purchase any product.
- Feedback: Can give review about any product to Customer Care Service at any time.
- Returns and Refunds: Customer can return products and expect refunds.

3.2 Non Functional Requirements

- Use encryption and authentication to keep card details safe during processing
- System should give quick response for card transactions within acceptable time limit.
- Users should get clear message if card is declined or if details are wrong.
- System should support payment through multiple card networks.
- Easy to use, efficient and accessible for merchants and users.
- Keep records of all transactions for future reference.

3.3 Interface Requirements

Hardware Requirements

- Processor - Atleast Intel Pentium IV 1.83
- Memory - Atleast 4GB RAM
- Hard Disk Drive - 100 GB+ free space for transaction storage.

Software Requirements
- Operating System: Windows / Linux / Android / iOS
- Database: Microsoft SQL Server Management Studio Express 2010

3.4 Performance Requirements
- The system shall accomodate high number of items and users without any fault.
- Responses to view information shall take no longer than 5 seconds to appear on the screen.
- The system shall process card transactions in less than 3 seconds under normal load.
- The system shall support atleast 500 users performing transactions without performance degradation.

4 Preliminary Schedule and Budget

4.1 Schedule.

Week	Key Goals	Tasks	Milestone Achieved.
Week 1	Requirement Gathering	Conduct client interviews, collect functional / non functional requirements, prepare initial use cases	Requirements document drafted.
Week 2	Requirement Finalization	Review and validate requirements with stakeholders freeze scope.	Approved SRS document
Week 3	System Design (High-Level)	Create system architecture diagram, define modules, database schema	High level design completed

Week	Key Goals	Tasks	Milestone Achieved
Week 4	System Design (Low Level)	Detailed class diagrams, data flow diagrams, interface definitions	Low level design completed
Week 5	Backend Development	Implement core database, API endpoints and server logic.	Core backend operational
Week 6	Frontend Development	Develop UI screens, integrate with backend API's	End to End worflow visible
Week 7	Integration and Unit Testing	Integrate all modules, perform unit and API testing.	Integration testing passed.
Week 8	System Testing	Conduct system testing, fix defects, performance tuning	stable system ready.
Week 9	UAT and Feedback	Deploy test system to client, gather feedback, make improvements.	Client sign off received.
Week 10	Deployment and Training	Deploy to production, provide user training and documentation	System Go-Live
Week. 11	Contingency / Buffer	Handle unforeseen delays fine tune based on real world usage	Project officially closed

Detailed Explanation of each week's Tasks:

- Week 1- Requirement Gathering- Collect user needs and document draft requirement
- Week 2- Review validate and freeze system requirements
- Week 3- Prepare architecture diagrams and select tech stack
- Week 4- Define database; modules, database and UI mockups
- Week 5- Implement database, APIs and business logic
- Week 6- Build UI screens and Integrate with backend
- Week 7- Combine modules and fix early defects
- Week 8- Test complete system for quality and performance
- Week 9- Deploy on staging and gather client feedback.
- Week 10- Go live and train end users.
- Week 11- Fix last issues and finally close the project.

4.2. Budget

Week	Key Goals	Estimated Cost(₹)
Week 1	Requirement Gathering	50,000
Week 2	Requirement Finalization	35000
Week 3	High Level Design	40000
Week 4	Low Level Design	35000
Week 5	Backend Development	125000
Week 6	Frontend Development	125000
Week 7	Integration and Unit Testing	70000
Week 8	System Testing and QA	80000

Week	Key Goals	Estimated Cost(₹)
Week 9	UAT and Feedback	40000
Week 10	Deployment and Training	50000
Week 11	Buffer and Closure	30000
	Total (Estimated)	680000

Notes:

- Costs include manpower effort only; infractructure, licenses and hosting are excluded
- Week 11 serves as a contingency buffer for unforeseen delays or bug fixes
- Costs are approximate and may vary slightly based on actual team size and hourly rates

This SRS provides a comprehensive overview of the credit card processing system, dealing its objectives components & interactions. It ensures that all stateholders share a clear understanding of the system's functionality and constraints. This document will serve as baseline for design, development and testing phases to deliver a reliable solution.

LIBRARY MANAGEMENT SYSTEM

Library Management System is one of the most common software development projects till date. A Library Management System is designed to efficiently manage operations of a library. The system aims to streamline and modernize the traditional library management process.

Problem Statement

The current manual library process is slow, error prone and does not scale with growing users. Tracking issued and returned books is cumbersome and often results in misplaced or duplicated records. Calculating fines and inforcing due dates is handled inconsistently and takes extra staff time. Searching the catalogue is time-consuming for both customers and staff, delaying the service. There is no quick-way to generate accurate reports on stock, circulation or user activity. Membership management, renewals and reservations are done on paper and are hard to audit. Multiple branch coordination is difficult because records are not centralized or synchronized. Therefore an automated Library Management system is required to streamline transactions, maintain records, and improve service efficiency.

1 Introduction

In the 21^{st} century, borrowing and returning books or viewing the available books in the library of the local university is currently done manually wherein the student has to go the Library and check the available books at the Library. Students check the list of books available and borrow the books if book is available to be borrowed or else it is waste of time to check for books in library. In case the book is

available, librarian checks the member id and allows the member to check out the book and librarian then updates the member database and also books' database manually which requires more time and manpower.
To overcome the problem of time, the idea of Online Library Management system that would be used by members who maybe students or faculty to check availability of books and borrow / return them comes up.

1.1 Purpose
This is the Software Requirements Specification for the Library Management system. The purpose of this document is to convey information about the application's requirements both functional and non functional. This document provides a description of the environment in which the application is expected to operate, a definition of the application's capabilities and specification of the application's requirements.
The document is intended to several groups of audience

- The SRS will be used by application designers who will use the information recorded here as basis for creating the system's design
- The client, that is the library manager is expected to review the document. SRS forms an agreement between clients and developer about functionality to be provided in application.
- Application maintainers will review the document to clarify their understanding regarding the application.
- Test planners use this document to derive plans and test cases.
- Project manager will use this document during planning and monitoring.

1.2 Scope

The Library Management System will be PC-base with a Local Area Network, allowing library users to search for books and library staff members to manage book inventory. Libraries consist of thousands of books, and with increase in number of readers, it needs to be well-organized. So, the system's obligation is to provide facility for book organization, monitoring them and controlling the transactions. It maintains details of books and members. It provides searching of available books. The main objective of the system is to simplify day to day process at library. It will be able to provide quick service in efficient manner. It will be able to remove drawbacks of larger customer information, physical files and catalog which were tough to manage. Secure Transaction, quick retrieval of information, ease of use, quick recovery of errors, fault tolerance are some of the benefits that the development team will be working on to achieve end user satisfaction.

1.3 Overview

A brief description of the content of each chapter is given below:

- Introduction: Provides an overview of the project. Summarizes major capabilities.
- General Description: Presents the environment in which the application is expected to operate, provides overview of system requirements, possible constraints on project, etc.
- Specific Requirements: Specification of requirements. Contains a complete description of application's requirements, functional and non functional.
- Preliminary Schedule and Budget: Schedule outlines the

planned timeline, key milestone and sequential tasks required to complete system. Budget provides an estimated review of costs required for developing the system.

2 General Description

2.1 Product Perspective

The main objective of this document is to illustrate the requirements of the project. This system is a replacement for the manual system of library management that makes tasks easy. This proposed system provides search facilities for books' names, ISBNs, author names and others to make readers' and staff members' lives easier.

2.2 Product Features / Functions

Administrators / Librarians

- They can update the book database by adding new book records, changing existing ones or removing outdated ones.
- They can manage user accounts, including adding, modifying or removing users.
- Librarians can send recall notifications and collect books from users when due dates are exceeded.
- They can generate reports that query book sign-out records or user records based on ISBN, user ID, title, time frame or overdue status.
- They are responsible for monitoring penalties for late pickups and maintaining accurate records of library transactions.

Users / Readers

- Users can create new library accounts & change account details
- They can borrow books by validating their library card, checking book limits and renewal limits.

- Users can check books in and out, and penalties will be applied automatically for late returns.
- They can search for books by title, author, ISBN or subject to find the desired book records.
- Users can also receive recall notifications when books need to be returned.

2.3 User Class and Characteristics

User Classes

- Library Manager
- Librarian
- Library User

Characteristics of User Classes

- Library Manager: Library managers are responsible for over-seeing the overall library operations and managing library staff. They have a good understanding of library workflows and good technical proficiency. Most library managers are familiar with older text-based terminal systems. To accomodate this user class, the system must provide a user interface with fewer input steps making it simple and easy to learn.
- Librarian: Librarians are responsible for day to day library operations such as check-in / check-out, managing user accounts, and handling bookrecords. They also have average technical proficiency and experience using text based terminals. System should offer a straightforward interface with minimal steps and a design that is intuitive, allowing librarians to perform frequent tasks quickly.
- Users / Readers: Library users consists of a diverse group, including, both younger and older generations. Younger users tend to learn adapt to new computer systems more easily.

while older users may prefer a system that resembles traditional workflows. Most users will not receive formal training. System must provide a graphical user interface that is easy to learn, along with system help and clear error messages for invalid inputs to ensure smooth usage.

2.4 Design and Implementation Constraints

- The information of all users must be stored in database that is accessible by Online Library Management System.
- The system database used should be an open source technology
- The system can potentially have hundreds of users. It is unrealistic to provide training to everyone. System should be designed for easy to use, providing help instructions, and appropriate error messages for invalid user inputs.
- Security is important to library operation. User should never break into the system and perform any modification.
- Reliability is vital to library operation. The system must not have any unscheduled downtime during library operation hours. It causes inconvenience to everyone in library.

3 Specific Requirements

3.1 Functional Requirements

- Library User Account Management: The system shall display user account details (ID, name position, privilege) and allow libraries to add, edit or remove user accounts. Notes can be attached to accounts, and all changes must be committed to database immediately.
- Book Borrowing and Returning: The system shall display all relevant user borrowing information (name, card number, expiry date, penalties). During check out or check-in it shall show book details (ISBN, title, due-date) and update database.

in real time. It shall enforce borrowing limits, verify privileges and allow librarians to process penalties and "library use only" books.

- Book Recall Management: The system shall list all borrowed copies of a recalled book, display waiting users in order of recall time, and track notification dates, arrival dates and pickup notifications. A check-in recall stamp shall be shown upon completion.
- Book Search: The system shall allow users to search books by title, keyword, ISBN, subject, author, category or language. Results shall be displayed in a sortable list, with the option to view detailed book information (including publisher, description and library location).
- Book Database Update: The system shall allow adding, editing or removing book records, including category, title, ISBN, publisher, description, location, purchase date and price. Deleted books must include a reason for deletion.
- Data Entry and Validation: The system shall support keyboard input, mouse selection, barcode scanning, and online calendar date selection. It shall validate all inputs and provide error messages to incorrect entries.

3.2 Non-Functional Requirements

- Use Secure login and authentication to protect user and librarian accounts.
- System should give quick response for search, issue and return operations within acceptable time.
- Users should get clear message if book is unavailable, reserved or request fails.
- System should support multiple user roles.
- Easy to use, accessible and efficient for library members.
- Keep record of all transactions (issues, returns, fines) for future

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Interface Requirements:

User Interface Requirements

- Users can login, register as new members, or recover forgotten passwords using a security question.
- Clear error messages are shown when login credentials are incorrect
- Search interface allows users to find books by title, category/ISBN
- Category view displays available book categories, and allows librarians to add, edit or delete categories.
- Control panel enables librarians to manage users, resources and lending options.
- Interface supports quick viewing of reports like books issued and returned within a time period.
- Stock verification and search are available through the interface.
- Design follows a standard, simple and customizable template.
- Login and logout modules have dedicated interfaces for secure access.

Hardware Requirements

Server:

- Operating System: Windows 7 or higher / LINUX
- RAM greater than 8 GB
- HDDs larger than 1000 GB.

Client side:

- Operating System: Windows 7 or higher / linux / MAC / Android
- RAM greater than 4 GB
- HDDs larger than 500 GB.

Software Requirements

- Any windows or Android OS
- A compatible runtime environment must be installed on the system to run the application.

Performance Requirements

- The system shall respond to check-in / check-out requests within 5 seconds and search requests within 9 seconds under normal load.
- The system shall maintain 99% reliability during library operating hours.
- Automatic recovery shall occur within 10 minutes in case of system failure, without user intervention.
- Meaningful error messages shall be displayed when invalid data is entered or when system is down.
- Only authorized users (librarians / managers) shall be allowed to access sensitive functions like database updates, report generation and resource management.
- Scheduled downtime shall not exceed more than 1 hour per day outside library working hours.
- System shall be compatible with library's network environment for smooth installation and use.

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4. Preliminary Schedule and Budget.

4.1 Schedule

Week	Key Goal	Tasks	Milestone
1-2	Requirement Collection and Documentation	Meet with librarians, students and admin staff to collect requirements. Document use cases, functional and non functional requirements.	Finalized SRS document
3	System Architecture and Planning	Design system architecture (3-tier), database schema (Books, Users, Transactions) and module responsibilities	Approved architecture and DB schema.
4-6	Development of Core Module	Develop frontend UI (book search, user login, checkout pages) and backend logic (issue, return, fine) in parallel	Working prototype with basic LMS features.
7	Integration	Integrate frontend and backend. Ensure search, issue, and return modules sync with real time database updates	Fully functional LMS with integrated modules.
8-9	Testing	Unit, integration testing and validation of edge cases Fix bugs and optimize performance	Bug free tested LMS ready for deployment
10	Deployment and Documentation	Deploy LMS to a test server or local network. Prepare user manual and admin guide	LMS successfully deployed and documented.

Week	Key Goal	Task	Milestone
11	Buffer / Improvement	Reserved time for unexpected delays, extra features / bugs	Final sign off from stateholders.

Detailed Explanation of Each Week's Tasks

- Week 1-2 Gather and document requirements from library staff & users
- Week 3: Design architecture, database and UI for the system.
- Week 4-6: Develop key modules like login, book management and transactions.
- Week 7: Integrate frontend and backend, ensuring smooth data flow.
- Week 8-9: Test for functional errors, performance and user experience issues.
- Week 10: Deploy system, train users and hand over documentation
- Week 11: Address final fixes and confirm system stability before closure.

4.2 Budget

Week	Task	Estimated Cost (₹)
1-2	Requirement collection + SRS preparation	50000
3	System architecture and design	75000
4-6	Development of modules (UI, backend)	3,00,000
7	Integration of Components	75,000
8-9	Testing and QA	1,00,000
10	Deployment and documentation	50,000
11	Bug fixing and final review	25,000

Total Estimated Cost = ₹6,75,000

Notes

- Cost estimates are assuming a small-medium scale LMS.
- Week 11 acts as buffer.
- Budget only includes manpower.

The Software Requirement Specification document clearly defines the functional and non functional requirements of the Library Management System. It outlines the system's purpose, user-classes, performance expectations, security measures and reliability goals. by providing a well structured description of features, interfaces and constraints, SRS serves as mutual agreement between stakeholders and developers.
Following this document will ensure that the Library Management System is user-friendly, secure, efficient and aligned with the needs of librarians, administrators, and patrons. It also provides a strong foundation for design, testing and future maintainance activities.

STOCK MAINTENANCE SYSTEM

The Stock Management System refers to the system and processes to manage the stock of organization with the involvement of Technology. This system can be used to store the details of the stock, stock maintainence, update the stock based on sales details, and generate daily or weekly report of stock. The stock management system is important to ensure quality control in businesses that handle transactions revolving around consumer goods. Without proper stock control, a large retail store may run out of stock on an important item. It is also means of tracking large shipment. An automated stock maintenance system helps to minimize errors while recording stock.

Problem Statement

The main purpose of the stock maintenance system is to manage the entire details of the stock system in an automated way. It must take care of sales Information of the company and must analyze the potential of the trade. The stocks which are purchased from various dealers and suppliers are stored in the store keeper and their entries are recorded into the database. The software system provides facilities for adding new item, removing an item, updating the stock, calculating the turn over, sales amount, total number of stocks. It also involves purchasing of stocks by the customers.

1 Introduction

A stock Maintenance System is a software solution designed to manage, monitor and control inventory efficiently within an organization. It keeps a digital record of every item purchased, stored, issued and returned ensuring that stock levels are always accurate and up to date. The system reduces the need for manual record-keeping which is often time-consuming and prone to errors.

Whenever a transaction takes place, such as adding new stock, selling items, or transferring goods between locations, the system automatically updates the records. This allows businesses to track the movement of goods in real time and generate useful reports like stock availability, reorder levels and consumption patterns.

By using a stock maintenance system, businesses can prevent stockouts, overstocking and wastage. It also ensures transparency by keeping a complete record of stock activities for audits and decision-making. These systems are commonly used in retail, warehouses, manufacturing and service sectors to streamline operations. Having accurate stock information, helps meet customer demand, cut costs, and boost overall efficiency.

1.1 Purpose

The entire process of stock maintanance is done in a manual manner considering the fact that the number of customers for purchase is increasing every year, a maintanance system is essential to meet the demand. The purpose of the SRS document is to present a detailed description of the stock maintenance system. It will explain the purpose and features of the software, the interfaces of the software, what the software will do, the constraints under which it must operate, and how software will react to external stimuli.

This document is intended both for end users and developers of the software.

Scope
This documento will cover the requirements of the Stock Maintenance system. This software will provide a geographical environment where users of the system will be able to perform various operations that are associated with storing, updating and retreiving Product information. The system provides a communication system between the customer and salesperson. The purpose is to guide the developers to select a design that is able to accomodate the full-scale application. The system will capture information about customer details, products and their quantities.

Overview
The document includes the general description of the system detailed functional and non functional requirements, design constraints and the preliminary schedule and budget required to design, develop and deploy the system.

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2 General Description

2.1 Product Perspective

The proposed stock maintenance system is an online platform designed to manage and track inventory efficiently. It will allow users to view available stock in real time, ensuring that customers can check product availability before purchasin-g them. The system will automatically update stock levels as items are sold or restocked, reducing manual errors. Additionally, it includes a customer feedback feature, enabling businesses to gather valuable insights about products and services. Overall, this system aims to improve customer satisfactio-n, streamline operations, and support better decision making.

2.2 Product Functions

For Naive Users (Customers)

- Users can create new accounts, login securely and update their profile information.
- They can view available stock of products in real time before placing an order.
- Users can request or reserve items if stock levels meet their needs.
- They can track the status of their requested orders, including packaging, dispatch and delivery.
- Users receive notifications about order status, stock availability or special offers.
- They can submit feedback or complaints regarding product availability, quality or details

For Warehouse Managers and Vendors:

- Managers and vendors can log in with credentials and role based access.
- Managers can add new stock records, update product details.

and remove obselete entries from the database.

- They can monitor inventory levels, set reorder thresholds, and receive automatic restock alerts
- Managers can generate detailed inventory reports based on product ID, date usage frequency or vendor performance
- Vendors can update shipment information, confirm goods received, and provide estimated arrival times.
- Managers can oversee order dispatch, track goods movement and coordinate with shipping vendors to avoid delays.
- They are responsible for maintaining stock records and ensuring timely replenishment to prevent shortages or overstocking.

2.3 User Class and Characteristics

User Classes

- Naive Users
- Warehouse Manager
- Inventory Management
- Shipping Vendor

Characteristics of User Classes

- Naive Users: They are end users such as common people, office goers and businesses who require products for daily use. They may not have technical knowledge so system must provide intuitive interface. They interact with system mainly to browse available products, order them and track the status of their order. They should be able to log-in securely, manage their personal profiles and view order history. Feedback and complaint submission must be easily accessible.
- Warehouse Managers: These users have administrative privileges and handle day to day warehouse operations. They are responsible for updating stock levels, adding new items, packaging

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and dispatching products. They use the system frequently and need efficient data entry and inventory update options. They also manage re-orders, generate inventory reports, and oversee all warehouse performance.

- Inventory Management Staff: These users monitor inventory levels closely and make decisions for re-ordering & procurement. They require access to detailed reports, alerts and stock analytics to plan purchases and prevent shortage/overstocking. They must have role-based access to edit stock levels and trigger purchase orders.
- Shipping Vendors: These are external partners responsible for pickup and delivery of orders. They need limited access to update shipment details, estimated delivery time, and provide order tracking information. They may not use the system very frequently, so their interface should be minimal and straightforward. Communication with warehouse manages must be smooth to ensure timely deliveries.

2.4 Design and Implementation Constraints

- Role based access control is implemented where each users and manages is assigned credentials for logging in.
- The administrator/manager has visibility over all user data whereas users cannot access administrator information.
- Credit card transactions undergo administrator verification and approval for authentication.
- Inventory and warehouse operations are performed manually.
- User feedback is systematically collected and stored for analysis and system improvement.

3 Specific Requirements

3.1 Functional Requirements

- The system shall be internet oriented and require an active online server, ensuring that all users can access the platform anytime and from any location with internet connectivity.
- The system shall store product details, customer details, supplier records, sales records, purchase details, and stock information securely in a centralized remote database for easy access and management.
- The system shall allow customers to login with their credentials and complete the purchasing process online, including selecting products, adding them to cart and confirming orders.
- The system shall allow the manager to view, filter, and print product, customer, supplier, sales, purchase, and stock details as needed for reporting and decision making purposes.
- The system shall provide functionality to generate and send purchase orders to suppliers electronically and receive digital invoices for proper record keeping.
- The system shall allow administration to manage and update product information, stock details, customer profiles and supplier records to maintain up-to date data in system

3.2 Non-Functional Requirements

External Interface Requirements

User Interface

- The system will have a simple and attractive Graphical User Interface for easy customer interaction.
- The introductory screen will display a welcome message and product advertisements.

- Customers can login, view product categories, check-details, and proceed with payment using card details.
- After purchase, the system will show "Thank You" message and update stock details immediately.

Hardware Interface
- HDD: 500 GB+ minimum
- RAM: 4 GB+ RAM
- A stable internet connection
- Monitors with good refresh rate for frequent stock updates.

Software Interface
- Windows 7+ / Linux / MAC operating system
- Stock Master version 3.0, MySQL, Microsoft Express SQL server Management studio.

Performance Requirements
- System must ensure smooth and ensure responsive performance particularly when browsing products, executing sales and processing purchases
- Database shall accomodate atleast 1 million records of stock, customer and suppliers.
- Software shall support concurrent access by multiple users without performance degradation
- System must ensure database protection through secure storage, backup and recovery mechanisms to prevent loss / corruption.
- A secure login mechanism to be implemented to prevent unauthorized access.
- The system must ensure that personal information remains confidential, preventing public from accessing sensitive data.

4. Preliminary Schedule and Budget:

Schedule.

Week	Key Goal	Task	Milestone Achieved
1	Stock and Supplier Requirement Study	Collect product categories, suppliers and customer details, stock movement rules.	Require Document signed off
2	System Design For Inventory	Design stock flow (inflow, outflow) supplier module, purchase/ sales cycle. Frontend design	Inventory Workflow finalized.
3.	Database Creation	Create tables for products, suppliers, stock levels, transactions	Database supports atleast a million records.
4.	Stock Management Module	Implement product add/ remove, quantity update, reorder alerts.	Core Stock Module Functional
5	Supplier and Customer Module Integration	Implement supplier registrat-ion, purchase order, customer details	Supplier and Customer Modules Completed.
6	Sales and Purchase Integration	Auto update stock after transaction. Generate invoice and billing screens.	Fully Integrated Sales-Purchase Flow.
7	Reporting and Deployment	Backend generates report APIs and the frontend shows low-stock alerts	Live system deployed

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Detailed Explanation of Each Week's Tasks.

- Week 1 - Covers product, supplier, stock movement and reporting needs.
- Week 2 - Ensures smooth worflow for stock management & usability
- Week 3 - Backend can handle CRUD operations for inventory records
- Week 4 - Users can add/edit product and monitor live stock levels.
- Week 5 - Supports vendor / customer management with full UI/DB sync.
- Week 6 - Stock auto adjusts after each purchase
- Week 7 - Fully deployed stock maintenance system.

Budget

Week	Task	Estimated Cost(₹)
1	Requirement Gathering & Analysis	40,000
2	System Architecture, UI/UX	75,000
3	Backend Development	1,50,000
4	Frontend Development	1,20,000
5	Inventory, Order, User Modules	65000
6	Testing, Q.A., Deployment	50,000
7	Documentation, Training	50,000
	Total (Estimated)	₹ 5,50,000

Notes

- Costs include manpower costs only. Infrastructure, licences, etc are excluded
- Costs may vary according to team size and hourly rates

This SRS defines a complete, secure and scalable stock

Maintainance System that streamlines inventory orders and warehouse operations. The solution uses a modern frontend, reliable backend and real time reporting to improve efficiency and support better decisions for businesses.

PASSPORT AUTOMATION SYSTEM

Passport Automation system is used in effective dispatch of passports to all applicants. This system adopts a comprehensive approach to minimize the manual work and schedule resources, time in cogent manner. The core of the system is to get online registration form (with details like name, address, etc) filled by the applicant whose testament is verified for its genuineness by the Passport Automation System with respect to already existing information in database. It aims at reducing complexities involved and improving efficiency in issue of passports.

Problem Statement

The current passport system is slow, manual and insecure, relying on in-person visits and government agents who may mishandle data and charge high fees. The proposed Passport Automated System uses an online portal for application registration, capturing personal data and documents digitally. It employs automated verification against a centralized database using secure algorithms to detect discrepencies. Verified applications are then securely transmitted to regional authorities for manual review. This reduces manual workload, and processing time while enhancing data security, through encryption and access controls.

1 Introduction

Passport Automation system is an interface between the Applicant and the Authority responsible for the Issue of Passport. It aims at improving the efficiency in the Issue of Passport and reduces the complexities involved in it to maximum possible extent.

1.1 Purpose

If the entire process of 'Issue of Passport' is done in a manual manner, then it would several months for the passport to reach the applicant. Considering the fact that the number of applicants for passport is increasing every year, an Automated System becomes essential to meet the demand. So this system uses several programming and database techniques to elucidate the work involved in the process. This document outlines the functional and non functional requirements of the proposed Passport Automation System. The specifications have been carefully reviewed and validated to ensure they meet the intended objectives and provide a reliable solution.

1.2 Scope

- The system provides an online interface to the user where they can fill in their personal details, and submit necessary documents.
- The authority concerned with the issue of passport can use this system to reduce his workload and process the application in a speedy manner.
- Provide a communication platform between applicant and administrator.
- Transfer data between Passport Issuing Authority and local Police for verification of applicant's information.

1.3 Overview.

SRS includes these sections: Overall description, Specific Requirements, Preliminary Schedule and Budget. Overall description describes the major role of system components and their interconnections. Specific requirements describe the roles and functions of actors. and also the project schedule defines the schedule and budget defines the estimated budget for cost-effective implementation.

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2 General Description

2.1 Product Perspective

The Passport Automation System acts an interface between the 'applicant' and the 'administrator'. This system tries to make the interface as simple as possible and at the same time not risk the data stored in. This minimises the time duration in which the user receives the passport. It ensures proper verificatio-n of users details through secure authentication before processing any request. System also maintains digital record of all applications, enabling easy tracking and transparency throughout the process.

2.2 Product Functions

- Secure Registration of Information: Applications can securely register and submit their personal details, through the system, ensuring data privacy.
- Appointment Scheduling: The system schedules applicant for manual document verification minimizing and ensuring orderly processing.
- Application Status Panel: Administrators can update and display passport application status, keeping applicants informed at every stage.
- SMS and Email Notification: Applicants receive timely updates via SMS and e-mail regarding their appointment, status and approvals.
- Report Generation and Authorization: The administrator can generate detailed reports and is the only authorized person to add eligible application information to database.

User Class and Characteristics

User Classes

- Applicant
- Administrator
- Police

Characteristic's of User Classes

- Applicant: These are the persons who desire to obtain the passport and submit their information to the database. They must provide accurate and complete details. The system ensures that their application is recorded safely for further processing.
- Administrators: They have certain privileges to update the passport status and approve the issuance of passports. They may also supervise a group of persons under them to verify documents and provide suggestions on whether or not to approve dispatch of passports.
- Police: They are people who, upon receiving information from the Passport Automation System, perform a personal verification of the applicant to check if they have any past or present criminal cases. They are vested with the authority to recommend declining an application to the Administrator if any discrepencies are found. They communicate through the Passport Automation System.

Design and Implementation Constraints

- Applicants require access to a computer to submit their information
- High priority is given to security to protect submitted data.
- Despite security measures, there remains a risk of intrusion in the online environment.
- Users must exercise caution and ensure accurate entry when submitting their information.

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3 Specific Requirements

3.1 Functional Requirements

- Passport Information: The system shall provide information about different types of passportsand guide users in selecting the appropriate type of passport suitable for their needs.
- Applying for Passport: The system shall allow users to apply for a passport by providing necessary details. It shall display a list of required certificates for each passport type.
- Document Uploading: The system shall enable users to upload the required documents online. It shall validate the file formats and completeness of uploaded documents.
- Payment: The system shall provide multiple payment modes for passport application fees and generate a payment confirmation after successful transaction.
- Verification Module: The system shall allow different departments to verify submitted certificates. The system shall update the user's application status after verification is complete.
- Authentication: The system shall authenticate users to ensure only valid users access the servicesand prevent unauthorized access by requiring credentials.
- Status Verification: The system shall allow users to check the current status of their passport application and also enable them to provide feedback and register complaints regarding services.

3.2 Non Functional Requirements

- The system shall encrypt all user-senstive data during storage & transmission.
- It shall support role based access control to protect information.
- It shall be compatible across web and mobile platforms.
- It shall provide secure backup and recovery mechanism for data integrity

External Interface Requirements

Hardware Requirements
- Processor : Intel Core i3 or above
- RAM: 4 GB minimum
- Storage : 500 GB HDD / SSD
- Network : Reliable internet connection (for online verification & uploads)
- Peripherals : Scanner / Printer

Software Requirements
- OS : Windows / Linux / MAC
- MySQL or Postgre SQL database
- Apache / Nginx for server
- SSL / TLS for encryption, role based authentication

Performance Requirements
- The system shall response to users actions in less than 5 seconds.
- The system shall support atleast 500 concurrent users without degradation
- The system shall scale to handle 2x increase in workload with less than 10% performance drop.
- The system shall restore data within 30 minutes after a failure.
- The system shall ensure 100% encryption for sensitive data in storage and transmission.

4 Preliminary Schedule and Budget.

Schedule

Week	Key Goal	Tasks	Milestone Achieved
1	Requirement Analysis	Collect functional and non functional requirement, prepare use-case diagram, ER models.	SRS document completed
2	System Design	Design system architecture, schema for applicant, admin, police, design UI wireframes.	Design Document Ready
3	Backend Setup	Setup servers, implement DB schema, configure authenticatio-n service	Database and authentication modules functional.
4	Frontend Development	Develop responsive UI, integrate with backend API.	User interface connected to backend.
5.	Integration of Modules	Connect applicant submission and admin verification	Core workflow integration achieved
6	Security Implementation	AES based encryption for/ sensitive data, SSL for secure communication	Secure data transfer and storage enabled.
7	Testing	Conduct unit, integration and load testing	Test report generated.
8	Deployment and Documentation	Deploy system on cloud/ Local server, prepare user manual and technical documentation	System live and documentation submitted.

Week	Key Goal	Task	Milestone Achieved
9	Contingency / Buffers	Fix unexpected bugs, optimize system performance handle missed deadline	Stable and optimized final release

Detailed Explanation of Weekly Task

- Week 1 - Gather and finalize requirements with technical diagrams.
- Week 2 - Design, architecture, database and UI mockups.
- Week 3 - Implement database and authentication system.
- Week 4 - Develop front end and link with backend API's
- Week 5 - Integrate applicant, admin and police modules
- Week 6 - Implement encryption and secure protocols
- Week 7 - Perform functional, load and security testing
- Week 8 - Deploy the system and finalize documentation
- Week 9 - Handle unexpected issues and ensure system is stable before delivery.

Budget

Week	Task	Estimated Cost (₹)
1	Requirement gathering and analysis	80000
2	Database schema and system architecture design	1,00,000
3	Backend services, API development	1,20,000
4.	UI prototype (frontend)	70.000
5.	DB integration and workflow logic	90,000
6.	Security Layer (Encryption, Authentication)	1,50,000
7	Testing (Unit, Integration, UAT)	80,000

Week	Task	Estimated Cost
8	Deployment setup and documentation	60,000
9	Contingency / bug fixing	50,000
	Total Estimated	₹8,00,000

Notes

- Cost covers only manpower - servers, cloud, hosting, licenses, etc, are not included.
- Week 9 acts as buffer
- Costs might vary depending on team size and hourly rates.

This Software Requirements Specification has clearly defined the functional and non-functional requirements, of the Passport Automation System. It outlines the purpose, user-reliability goals, performance, security measures by providing a well structured description of features, interfaces and constraint-s. SRS serves as a mutual agreement between the developer-s and stakeholders. Following this document will ensure that the Passport Automation System is user-friendly, secure, efficient and aligned with needs of its related people. It also provides strong foundation for design, development, testing and future maintenance activities.